

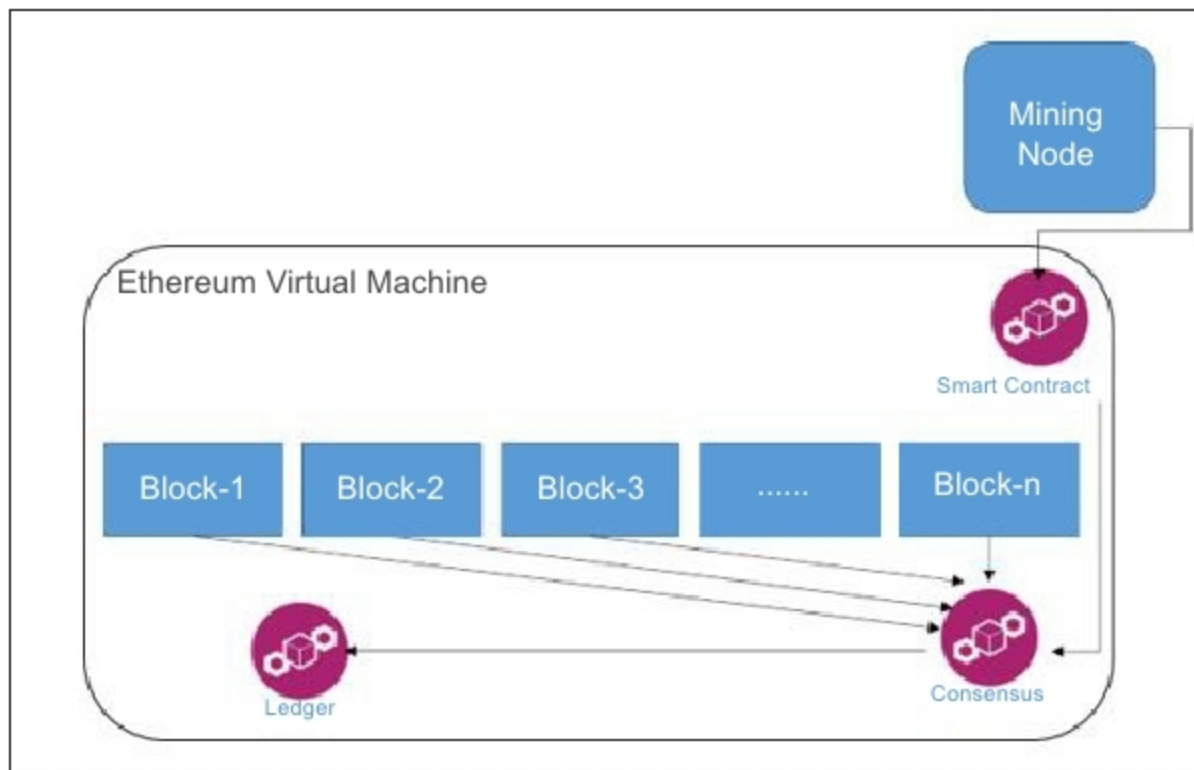


Figure 2: Ethereum architecture

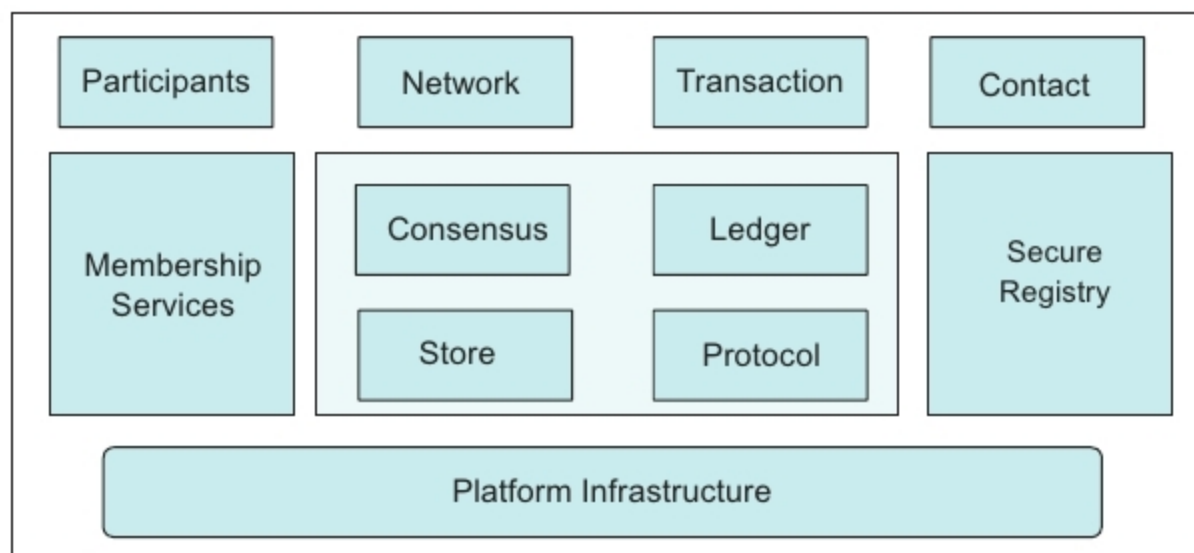


Figure 3: OpenChain's platform architecture

external accounts, the state comprises the account balance while for contract accounts, the state is defined by the memory storage and balance.

How Ethereum is different

- Ethereum offers greater transparency as compared to Hyperledger.
- The flexibility of the platform allows any developer to create applications using an inbuilt programming language.
- Having 'Ether' as a built-in cryptocurrency provides a competitive edge to Ethereum applications over Hyperledger and Corda in use cases that require cryptocurrency.

Key highlights

- *Activity:* Highly Active on GitHub
- *Type of network:* Public, smart contract based
- *Pricing:* For transactions and for computational services
- *Supported languages:* Python, GO, C++
- *GitHub repo:* pyethereum (Python); gopethereum (GoLang); cpp-ethereum (C++)

OpenChain

OpenChain is an open source distributed ledger technology. It is suited for organisations wishing to issue and manage digital assets in a robust, secure and scalable manner.

Figure 3 explains the architecture blocks of the OpenChain platform, which contains multi-layer components for consensus, ledger and platform services, and supports membership services to handle workflow and permissioned transaction management.

- Anyone can spin up a new OpenChain instance within seconds.
- The administrator of an OpenChain instance defines the rules of the ledger.
- End users can exchange value on the ledger according to those rules.
- Every transaction on the ledger is digitally signed, like with Bitcoin.

How OpenChain is different

- Tokens on OpenChain can be pegged to Bitcoin, making it a side chain.
- Has smart contract modules and unified API.
- Assigns aliases to users instead of using base-58 addresses.
- Multiple levels of control and an hierarchical account system allowing permissions to be at any level.
- Ability to have multiple OpenChain instances replicating from each other.

Key highlights

- *Activity:* Medium active on GitHub
- *Type of ledger:* Distributed ledger technology and is private
- *Pricing:* Open source
- *Supported languages:* JavaScript
- *GitHub repo:* Openchain-js

BigChainDB

BigChainDB is built from a federation of enterprise-ready database nodes, such as MongoDB instances, which store immutable information about assets in a synchronised manner.

BigChainDB is a MongoDB database that uses Tendermint to obtain its blockchain features. A BigChainDB network may be public, private or permissioned, according to the access